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Paradise Lost? Estimating the Economic Impact of Hawaii's Mandatory Quarantine using Synthetic Historical Controls

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Summary This article estimates the causal impact of Hawaii's March 2020 mandatory quarantine on its tourism industry and labor market using the Synthetic Historical Control method. Analyzing visitor days, hotel occupancy, accommodations employment, and the Coincident Economic Activity Index (all in annualized year-over-year growth rates), the study reveals substantial negative impacts from the quarantine. Observed outcomes fall significantly outside 90% conformal out-of-sample prediction intervals, indicating a clear treatment effect. Placebo tests further validate the model's internal credibility. This research demonstrates the method's utility and informs the broader debate on economic impacts of anti-COVID policy.

Keywords: *Synthetic Controls, Policy Evaluation, COVID-19*

1. INTRODUCTION

Hawaii's response to the COVID-19 pandemic ranks among the most extreme economic interventions enacted by any U.S. state. This paper studies the short-run economic consequences of its March 2020 mandatory quarantine (Yan et al., 2022). Unlike most regions that experienced economic contraction from COVID-19 through voluntary distancing, fear, or uneven mandates, Hawaii implemented one of the few deliberate and coordinated state-level shutdown in the United States. This policy was aimed not merely at mitigating the virus, but at suppressing travel and commerce altogether. On March 26, 2020, the state imposed a mandatory 14-day quarantine on all arrivals, effectively sealing its borders.

This paper addresses a core policy question: What are the economic costs of decisive early action in the face of an uncertain crisis? While much has been written about the epidemiological and social effects of COVID-19 interventions, less is known about their economic toll. This is especially true in cases where policymakers opted for maximal containment. The most famous case was Wuhan's January 2020 lockdown, as studied by Ke and Hsiao (2022). However, work outside of this context is rare. Understanding these costs is essential not just for pandemic preparedness, but for future crises, such as climate shocks or geopolitical disruptions, where early intervention may again carry significant short-run economic risks in exchange for long-term societal benefits (Newman and Noy, 2023).

Hawaii offers a compelling empirical case. Its geographic isolation and economic dependence on tourism allowed it to pursue a form of near-total border control that was logistically and politically infeasible in most mainland states. As such, Hawaii offers a rare glimpse into how a near-zero-COVID policy—one that *actually* shut down travel and commerce—might play out within a U.S. federal system. While other states (such as Texas and Georgia) issued stay-at-home orders or imposed limited business closures, few enforced them strictly, and most reopened early under political and economic pressure. Hawaii, by contrast, implemented a full-scale quarantine and border closure, a level of intervention unmatched by any other U.S. jurisdiction. In that sense, its strategy might be called "The Great Isolation with Aloha characteristics", even if it was not quite as extreme as Wuhan.

Despite its policy distinctiveness and the strength of its intervention relative to the mainland, Hawaii has been largely excluded from existing empirical evaluations. This is partly due to the context of the policy, given the lack of suitable control units to draw comparisons from. Many of the most pressing policy questions in public policy and economics concern one-time, systemic shocks: What would the United States' economy have looked like absent the American revolution? What would have happened to unemployment in Great Britain without Brexit? In these cases, credible untreated units often do not exist, making standard causal inference tools, like difference-in-differences (DiD) and synthetic control (SCM), difficult or impossible to apply.

To compensate for this, I employ the Synthetic Historical Control (SHC) method by Chen et al. (2024). SHC avoids the need for external control units by leveraging the treated unit's own pre-treatment dynamics, allowing credible counterfactual construction even in globally shocked contexts. Empirically, I find that Hawaii experienced substantial—but mostly short-lived—economic losses across the tourism and labor sectors, easing by early 2021.

The remainder of the paper proceeds as follows. Section 2 reviews prior research on early pandemic interventions and outlines key methodological challenges in studying one-off, jurisdiction-wide policies. Section 3 presents the SHC framework and its application in this setting. Section 4 describes the data. Section 5 reports the empirical results. Section 6 discusses broader implications for public policy and crisis response.

2. BACKGROUND

Hawaii implemented some of the earliest and most aggressive COVID-19 containment policies in the United States. On March 4, 2020, Governor David Ige issued an emergency proclamation officially recognizing the virus as a statewide threat (Maui Now Staff, 2020). Within weeks, the state began taking unprecedented measures alongside states like California (Friedson et al., 2021). On March 17, Governor Ige publicly urged all visitors to suspend their trips to Hawaii for 30 days, anticipating substantial economic damage to the state's tourism-dependent economy. *"We do know there will be significant impact,"* he stated, though no concrete estimates were available at the time (Nakaso, 2020). A sweeping stay-at-home order followed on March 25, banning non-essential activities, closing non-critical businesses, and effectively shutting down day-to-day life and travel across the islands (HNN Staff, 2020). These actions were implemented alongside a mandatory 14-day quarantine on all arrivals, passed the same week, which further intensified the shutdown and isolated the state from both domestic and international travel. Despite the unprecedented scope of these interventions, no formal quantitative analysis has yet estimated the short-run economic effects of Hawaii's approach. This paper provides such an assessment.

2.1. Literature Review

The onset of the COVID-19 pandemic spurred a global wave of NPIs, with governments acting swiftly to limit viral transmission. Much of the early research focused on the public health rationale behind these policies—particularly the value of acting early to prevent exponential growth in cases. Friedson et al. (2021), for instance, study California's shelter-in-place order and argue that its primary purpose was to delay the pandemic's peak, buying time for hospitals to prepare. They emphasize the importance of California's timing—implementing the policy while infection growth was still modest—as a key to its effectiveness.

Other studies reinforce the value of early, aggressive containment. Using SCM, Yang (2021) analyzes Wuhan's lockdown and estimate substantial reductions in population movement and COVID-19 transmission, emphasizing the relevance of their findings for international public health policy. Cerqueti et al. (2021), studying Italy's decision to close schools, stress that timing is critical when dealing with exponential contagion. They also note a political economy challenge: if containment is successful, the intervention may appear excessive in retrospect, raising political costs for early action. Bayat et al. (2020) offer a quantitative example, estimating that deaths in New York could have been reduced by over 80% had lockdown been enacted ten days earlier. Outside of China and Europe, similar insights emerge. Sobiech Pellegrini (2022) and Li and Shankar (2023) document similar patterns in Brazil and Texas, finding that premature reopenings by state governments led to sustained spikes in cases.

Collectively, these studies underscore a key lesson: early interventions can mitigate public health disasters, but often at economic cost. This tradeoff between health and the economy became central to both policymaking and public discourse. Osman et al. (2022) document how opponents of lockdowns, particularly in conservative circles, criticized them as economically inefficient and ineffective. Gibson and Sun (2023) evaluate statewide stay-at-home orders using synthetic control, focusing on new jobless claims as an economic indicator of interest. Lee and Yang (2022) examine the labor market impacts of COVID-19 in South Korea, highlighting the economic consequences of NPIs. Ke and Hsiao (2022) study the economic impact of Hubei's lockdown. They show that while Hubei's lockdown caused a sharp downturn in the short term, the effects were short-lived—suggesting that delay or inaction might have led to more lasting harm.

Other work has assessed broader welfare outcomes of interventions. Cole et al. (2020) show how the Wuhan lockdown reduced air pollution and associated mortality using augmented SCM. Oliu-Barton et al. (2022) study vaccine mandates in Europe and estimate that increased vaccine uptake not only averted deaths but also prevented GDP losses—suggesting that public health policy can have broader economic upside when well-designed. Despite the breadth of this literature, Hawaii's case remains notably absent. Yet it represents perhaps the clearest U.S. example of an early, and most aggressive intervention. This paper contributes by bringing Hawaii into the empirical record and by offering credible estimates of the short-run economic costs of a "zero-COVID'' style border (Yuan, 2022).

While much has been learned about the effects of pandemic interventions, measuring their economic impacts remains challenging. Most studies evaluating anti-COVID policies rely on difference-in-differences (DiD) or synthetic control methods (SCM). Where cross-sectional variation in policy exists—e.g., some counties or states implementing lockdowns while others do not—SCM and panel methods have been used productively (e.g., Yang, 2021; Ke and Hsiao, 2023; Gong et al., 2024). However, these designs depend on the availability of unaffected or minimally affected units to serve as comparisons. As noted by Callaway and Li (2023) and Bilgel (2022), this and other critical assumptions often break down in the context of a global pandemic, where nearly all regions experienced some form of shock.

These limitations apply especially in Hawaii's case. The state is economically and geographically distinct, and every other U.S. state (as well as comparable island economies) faced their own pandemic-related disruptions. There is no unaffected or even roughly comparable unit that can serve as a credible counterfactual for Hawaii's tourism and labor markets. In such cases, standard cross-sectional methods fail. To my knowledge, the only study examining Hawaii's economic response to the pandemic is Bond-Smith and Fuleky (2022), which provides valuable descriptive analysis but does not construct counterfactual estimates.

To address this challenge, I apply the SHC method developed by Chen et al. (2024). SHC circumvents the need for external control units by leveraging the treated unit's own pre-intervention history. Rather than borrowing information from across space, it borrows strength across time, enabling credible inference in settings where conventional comparative approaches are infeasible—precisely the situation posed by Hawaii's quarantine.

3. METHODOLOGY

We know what happened given that Hawaii mandated two-week quarantines. We do not know what would have happened had they not done this. Given its novelty and radical deviation from SCM, this section summarizes the SHC method by Chen et al. (2024). For a comprehensive treatment of the method, including the theoretical proofs and asymptotic properties, readers are referred to Chen et al. (2024).

Notation Let $\mathbb{R}$ denote the set of real numbers, and let $\|\cdot\|_1$ denote the usual ℓ_1 vector norm. Throughout, I denote sets using calligraphic letters (e.g., $\mathcal{T}, \mathcal{S}, \mathcal{N}$) for clarity. Scalars are represented using plain lowercase letters (e.g., h, t, n, m), and matrices are denoted by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{W}, \mathbf{Y}$). Given a vector $\mathbf{x} = (x_1, x_2, \ldots, x_T)^\top \in \mathbb{R}^T$ and an index set $\mathcal{I} \subseteq \{1, \ldots, T\}$, I write $\mathbf{x}_\mathcal{I} := (x_t)_{t\in\mathcal{I}} \in \mathbb{R}^{|\mathcal{I}|}$ to denote the subvector of $\mathbf{x}$ corresponding to indices in $\mathcal{I}$, preserving their original order.

Let $\mathcal{T} = \{1, 2, \ldots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \ldots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period. To reconstruct the evaluation window, we compare it against earlier segments of the treated unit's own pre-treatment trajectory.

Define the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$ as a collection of N overlapping length-m subvectors extracted from the smoothed pre-treatment series. Each column is a contiguous subvector of the form $\widetilde{\mathbf{y}}_{[i,i+m-1]}$, with i ranging over eligible start points in $\mathcal{T}_1$ that precede the evaluation window. The precise construction is given in Section 3.2.

3.1. Synthetic Historical Control Method

Potential Outcomes Following the Rubin potential outcomes framework, let y_t^1 and y_t^0 denote the potential outcomes for the treated unit at time $t \in \mathcal{T}$, corresponding to the outcomes under treatment and no treatment, respectively. The observed outcome at time t is $y_t = d_t y_t^1 + (1 - d_t) y_t^0$, where $d_t = \mathbf{1}\{t > T_0\}$. The fundamental problem of causal inference is that for $t > T_0$, we observe only the treated potential outcome $y_t^1 = y_t$, while the untreated potential outcome y_t^0 remains unobserved. Let the SHC estimator for y_t^0 be denoted $\widehat{y}_t^0 := y_t^{\text{SHC}}$ for all $t \in \mathcal{T}_2$. Define the observed post-treatment outcome vector as $\mathbf{y}_{\text{post}} := (y_t^1)_{t \in \mathcal{T}_2}$ and the estimated counterfactual vector as $\mathbf{y}_{\text{post}}^0 := (\widehat{y}_t^0)_{t \in \mathcal{T}_2}$. The out-of-sample treatment effect is

$$\alpha_t := y_t^1 - y_t^0, \quad \text{for } t \in \mathcal{T}_2, \tag{3.1}$$

with estimator $\widehat{\alpha}_t := y_t - \widehat{y}_t^0$. Stacking over $t \in \mathcal{T}_2$ yields:

$$\widehat{\boldsymbol{\alpha}} := \mathbf{y}_{\text{post}} - \mathbf{y}_{\text{post}}^0. \tag{3.2}$$

The average treatment effect on the treated (ATT) is then given by:

$$\widehat{\text{ATT}} := \frac{1}{n} \sum_{t=T_0+1}^{T_0+n} \left(y_t - \widehat{y}_t^0 \right) = \frac{1}{n} \mathbf{1}^\top \widehat{\boldsymbol{\alpha}}, \tag{3.3}$$

where $\mathbf{1} \in \mathbb{R}^n$ is a vector of ones.

3.2. The Model

The SHC method introduced by Chen et al. (2024) constructs a counterfactual using the treated unit's own pre-treatment time series.[1] SHC requires a few assumptions: Assumption 3.1, Assumption 3.2, and Assumption 3.3.

Assumption 3.1. (Error Properties) *The error term satisfies* $\mathbb{E}[\varepsilon_t] = 0$ *and* $\mathbb{E}[\varepsilon_t^2] < \infty$ *for all* $t \in \mathcal{T}$.

[1] SCM estimates the counterfactual by solving:

$$\mathbf{w}^* = \underset{\mathbf{w} \in \Delta^{N_0}}{\text{argmin}} \, \|\mathbf{y}_1 - \mathbf{Y}_0 \mathbf{w}\|_2^2, \quad \forall t \in \mathcal{T}_1, \tag{3.4}$$

where $\mathbf{Y}_0 \in \mathbb{R}^{T_0 \times N_0}$ is the donor matrix of unexposed units and $\Delta^{N_0} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^{N_0} : \|\mathbf{w}\|_1 = 1 \right\}$ is the probability simplex.

Assumption 3.1 says unmodeled fluctuations or noise in the outcome are random.

Assumption 3.2. (Latent Component Smoothness and Matching Condition) *The latent trend $\widetilde{y}(t)$ is $(H+1)$ times differentiable for some integer $H \geq 3$, with derivative bounded as $\left|\widetilde{y}^{(H+1)}(t)\right| < \varepsilon$ uniformly over $t \in \mathcal{T}_1$. Additionally, $\exists \mathbf{w}^* \in \Delta^{N-1}$ such that $\mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*$.*

Assumption 3.2 allows us to approximate the post-treatment latent component using a local linear extrapolation of pre-treatment values and posits that perfect match exists within the donor pool for the evaluation window, much as Abadie et al. (2010) assume about donor pool units matching to the treated unit.

Assumption 3.3. (Feasibility) *There exists a weight vector $\mathbf{w} \in \Delta^{N-1}$ such that $\widehat{\mathbf{y}}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}$, with $\|\mathbf{w}\|_0 \leq s \ll N$. Let $\widetilde{\mathbf{Y}}^{(s)}_{pre}$ denote the submatrix formed by the active columns of $\widetilde{\mathbf{Y}}_{pre}$ indexed by* $\operatorname{supp}(\mathbf{w})$*; then its Gram matrix satisfies $\left(\widetilde{\mathbf{Y}}^{(s)}_{pre}\right)^\top \widetilde{\mathbf{Y}}^{(s)}_{pre} \succ 0$.*

Chen et al. (2024) note that Assumption 3.3 is the sample-variant of Assumption 3.2. It says that the SHC weights are identifiable, sparse, and numerically stable, similar to the assumptions described by Abadie (2021).

The first step of the SHC method smooths $\mathbf{y}_{\text{pre}}$ using local linear regression, where we fit the regression model:

$$y_j \approx a_t + b_t(j-t), \quad \text{for } j = 1, \ldots, T_0, \tag{3.5}$$

with local intercept $a_t \in \mathbb{R}$ and slope $b_t \in \mathbb{R}$. Define the centered time vector $\mathbf{s}_t = (1-t, 2-t, \ldots, T_0-t)^\top \in \mathbb{R}^{T_0}$, and the design matrix $\mathbf{X}_t = [\mathbf{1}\ \mathbf{s}_t] \in \mathbb{R}^{T_0 \times 2}$. The diagonal kernel weight matrix is:

$$\mathbf{W}_t = \operatorname{diag}(w_{1t}, w_{2t}, \ldots, w_{T_0 t}), \quad w_{jt} \propto \exp\left(-\frac{1}{2}\left(\frac{j-t}{h}\right)^2\right), \tag{3.6}$$

where $(h > 0$ is a bandwidth parameter governing smoothness. The local coefficients $\boldsymbol{\beta}_t = (a_t, b_t)^\top$ solve quadratic programming problem that is strictly convex

$$\boldsymbol{\beta}_t = \underset{\boldsymbol{\beta} \in \mathbb{R}^2}{\operatorname{argmin}} \left\| \mathbf{W}_t^{1/2} \left(\mathbf{y}_{\text{pre}} - \mathbf{X}_t \boldsymbol{\beta}\right) \right\|_2^2. \tag{3.7}$$

Fortunately, this optimization admits a closed-form solution for the coefficients:

$$\boldsymbol{\beta}_t = \left(\mathbf{X}_t^\top \mathbf{W}_t \mathbf{X}_t\right)^{-1} \mathbf{X}_t^\top \mathbf{W}_t \mathbf{y}_{\text{pre}}. \tag{3.8}$$

The smoothed outcome at time t is $\widetilde{y}_t = a_t = \mathbf{e}_1^\top \boldsymbol{\beta}_t$, where $\mathbf{e}_1 = (1,0)^\top$. Let $\widetilde{\mathbf{y}} = (\widetilde{y}_1, \ldots, \widetilde{y}_{T_0})^\top \in \mathbb{R}^{T_0}$ denote the full smoothed series. After we have smoothed over the pre-treatment period before the evaluation period, we build our donor matrix of historical control segments/units.

In the second step, SHC constructs $N = T_0 - m - n + 1$ overlapping temporal segments of length m. Each segment $\widetilde{\mathbf{y}}_{[i,i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}} := \begin{bmatrix}\widetilde{\mathbf{y}}_{[1,m]} & \widetilde{\mathbf{y}}_{[2,m+1]} & \cdots & \widetilde{\mathbf{y}}_{[N,N+m-1]}\end{bmatrix} \in \mathbb{R}^{m\times N}$.

EXAMPLE 3.1. (CONSTRUCTING $\widetilde{\mathbf{Y}}_{\text{PRE}}$) *Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months in order to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is: $N = T_0 - m - n + 1 = 60 - 12 - 6 + 1 = 43$. Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.*

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w}\in\mathbb{R}^N}{\operatorname{argmin}} \quad & \left\|\widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}\mathbf{w}\right\|_2^2 + \varsigma \left\|\mathbf{C}_0^\top \mathbf{w}\right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1 \end{aligned} \tag{3.9}$$

Chen et al. (2024) note that in the general case when we have more donor units than we do the length of the segments ($N > m$), the optimization becomes an ill-posed problem. The regularization term stabilizes the optimization by penalizing weight allocations along directions of low variance in the donor pool, reducing sensitivity to collinear segments. Formally, $\mathbf{C}_0$ contains the eigenvectors of the Grammian matrix $\mathbf{G} \in \mathbb{R}^{N\times N}$. $\mathbf{C}_0$ corresponds to the eigenvalues less than $\tau > 0$. These directions capture low-variance combinations of donor segments, and the penalty shrinks weights in those directions.[2]. The corresponding weight vector $\mathbf{w}^*$ is used to compute the counterfactual by applying it to the shifted donor segments:

$$\widetilde{\mathbf{Y}}_{\text{post}} = \begin{bmatrix}\widetilde{\mathbf{y}}_{[1+m,m+n]} & \widetilde{\mathbf{y}}_{[2+m,m+n+1]} & \cdots & \widetilde{\mathbf{y}}_{[N+m,N+m+n-1]}\end{bmatrix} \in \mathbb{R}^{n\times N}. \tag{3.10}$$

This yields in-sample and out-of-sample predictions for the SHC estimator. The full counterfactual vector is then: $\mathbf{y}^0 = \widetilde{\mathbf{Y}}\mathbf{w}^* \in \mathbb{R}^{m+n}$.

Bandwidth Selection via Cross-Validation To select the bandwidth parameter h used in local linear smoothing of $\mathbf{y}_{\text{pre}}$, I implement a *leave-one-out cross-validation* (LOOCV) procedure over a grid of candidate values $h \in [h_{\min}, h_{\max}] \subset \mathbb{R}_{>0}$. This procedure chooses the bandwidth that minimizes out-of-sample prediction error for the smoothed pre-treatment series $\widetilde{\mathbf{y}} \in \mathbb{R}^{T_0}$, as defined in Equation~(3.8).

For each candidate h, and for each time point $t \in \mathcal{T}_1$, define Gaussian kernel weights

$$w_{\tau t}^{(h)} = \exp\left(-\frac{1}{2}\left(\frac{\tau - t}{h}\right)^2\right), \quad \tau \in \mathcal{T}_1 \setminus \{t\}, \tag{3.11}$$

used to construct the diagonal matrix $\mathbf{W}_t^{(-t)} \in \mathbb{R}^{(T_0-1)\times(T_0-1)}$, omitting the index t. Let $\mathbf{X}_t^{(-t)} \in \mathbb{R}^{(T_0-1)\times 2}$ and $\mathbf{y}_{\text{pre}}^{(-t)} \in \mathbb{R}^{T_0-1}$ denote the design matrix and response vector with time t excluded.

Then the local linear coefficients $\boldsymbol{\beta}_t^{(-t)} = (a_t^{(-t)}, b_t^{(-t)})^\top$ are obtained by solving Equation~(3.7). The leave-one-out smoothed value at time t is then:

$$\widetilde{y}_t^{(-t)} = a_t^{(-t)} = \mathbf{e}_1^\top \boldsymbol{\beta}_t^{(-t)}, \quad \text{where } \mathbf{e}_1 = (1,0)^\top. \tag{3.12}$$

[2]The ς (sigma) term is a small positive constant used to stabilize the solution in cases of near multicollinearity.

To pick the ideal bandwidth, we minimize:

$$h^* = \underset{h \in [h_{\min}, h_{\max}]}{\operatorname{argmin}} \left\{ \mathrm{CV}(h) := \frac{1}{T_0} \sum_{t=1}^{T_0} \left(y_t - \widetilde{y}_t^{(-t)} \right)^2 \right\}. \tag{3.13}$$

Once selected, the optimal bandwidth h^* is used to construct the full smoothed pre-treatment series.

Control Group Selection Chen et al. (2024) note that we should have a parsimonious approach to which historical controls we use, rather than choosing all of them. In their examples, SHC typically requires less than 28 controls. They advocate for a forward selection procedure to choose the optimal control group. I implement a slightly modified variant of this method to choose the ideal number of historical controls. Let $\mathcal{N} = \{1, 2, \ldots, N\}$ index the full set of donor segments in the pre-treatment matrix $\widetilde{\mathbf{Y}}_{\text{pre}} \in \mathbb{R}^{m \times N}$. At each step the algorithm selects the donor index $i_j \in \mathcal{N} \setminus \mathcal{S}_{j-1}$ that, when added to the current donor set $\mathcal{S}_{j-1}$, minimizes the in-sample mean squared error (MSE) between the evaluation window and in-sample fit:

$$\mathcal{S}_j = \mathcal{S}_{j-1} \cup \left\{ \underset{i \in \mathcal{N} \setminus \mathcal{S}_{j-1}}{\operatorname{argmin}} \left\| \widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S}_{j-1} \cup \{i\})} \mathbf{w}^{(j)} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\}, \quad \mathcal{S}_0 = \emptyset, \tag{3.14}$$

where $\widetilde{\mathbf{Y}}_{\text{pre}}^{(\mathcal{S})}$ denotes the submatrix of $\widetilde{\mathbf{Y}}_{\text{pre}}$ with columns indexed by $\mathcal{S}$, and the weights $\mathbf{w}^{(j)} \in \Delta^j$ are the optimal simplex-constrained weights for the selected donor set $\mathcal{S}_j$. This procedure greedily expands the donor set one segment at a time, always choosing the segment that yields the greatest marginal reduction in in-sample error.

To avoid overfitting, Chen et al. (2024) manually choose the number of donors the forward selection procedure should choose. In their case, they experiment with the single best historical segment (which rarely does well), the best 15, 20, and 27 historical control units. However, this is a subjective decision that will yield different results depending on what we select. To mitigate researcher degrees of freedom, I view the number of historical segments as a tuning parameter. I employ a forward selection-based algorithm to choose the number of controls. For my implementation of SHC, I employ an early stopping rule based on a modified Bayesian information criterion (BIC), borrowing inspiration from Shi and Huang (2023). For each donor set size j, I compute:

$$\mathrm{BIC}(j) = m \cdot \log\left(\mathrm{MSE}_j\right) + \lambda j, \tag{3.15}$$

where MSE_j is the in-sample error using donor set $\mathcal{S}_j$, and $\lambda = \log(m)$ penalizes the inclusion of too many donors. The selection procedure terminates when $\mathrm{BIC}(j)$ increases for two consecutive steps. The final donor set $\mathcal{S}^*$ is then taken to be $\mathcal{S}_{j^*}$, where:

$$j^* = \underset{j}{\operatorname{argmin}}\ \mathrm{BIC}(j). \tag{3.16}$$

This approach adaptively selects a sparse donor subset that closely matches the treated unit's pre-intervention trajectory. The selected donor set $\mathcal{S}^*$ is used to compute the SHC model predictions. I would like to mention that this forward selection method holds much in common with recently proposed methods from the literature. For example, Shi and Huang (2023) propose the forward selected panel data approach which selects the comparison group for the panel data approach by Hsiao et al. (2012) using an aggressive forward selection procedure. Li (2024), correcting for overfitting in the former, proposed the forward Difference-in-Differences estimator, where a forward selection approach is used to select the donor pool for the DiD model. Similar advances have been made SCM, as developed by Cerulli (2024), where a similar selection mechanism is used to estimate the treatment effect. It would be interesting, but is far outside the scope of this work, to compare these estimators to one another and see when their in-sample fits and point estimates agree with one another.

Inference Quantifying uncertainty around the counterfactual predictions is essential for valid inference. Chen et al. (2024) do not provide any inferential method to use aside from in-time placebos. Following recent work in post-modeling inference by Cattaneo et al. (2025), I construct agnostic conformal prediction intervals around the counterfactual series. These intervals are distribution-free, require no parametric assumptions, and are valid under exchangeability of residuals between the pre- and post-treatment periods (or, good in-sample fit).

Let $\mathbf{y}^0_{\text{eval}} \in \mathbb{R}^m$ denote the SHC-fitted values during the evaluation period and define the residuals as:

$$\boldsymbol{\varepsilon}_{\text{eval}} := \mathbf{y}_{\text{eval}} - \mathbf{y}^0_{\text{eval}}. \tag{3.17}$$

I then compute the sample mean $\bar{\varepsilon}$ and variance $\widehat{\sigma}^2$ of these residuals:

$$\bar{\varepsilon} = m^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} \varepsilon_t, \quad \widehat{\sigma}^2 = (m-1)^{-1} \sum_{t \in \mathcal{T}_{\text{eval}}} (\varepsilon_t - \bar{\varepsilon})^2. \tag{3.18}$$

To construct prediction intervals for the post-treatment counterfactual $\mathbf{y}^0_{\text{post}} \in \mathbb{R}^n$, I assume the residuals are sub-Gaussian and apply a conformal bound based on Hoeffding-type concentration inequalities. For a miscoverage rate $\alpha \in (0,1)$, the interval half-width is given by:

$$c_\alpha = \sqrt{2\widehat{\sigma}^2 \log\left(\frac{2}{\alpha}\right)}. \tag{3.19}$$

Then, the agnostic $(1-\alpha)$ prediction interval for each $t \in \mathcal{T}_2$ is given by:

$$\left[\widehat{y}^0_t + \bar{\varepsilon} - c_\alpha,\ \widehat{y}^0_t + \bar{\varepsilon} + c_\alpha\right]. \tag{3.20}$$

Stacking these yields lower and upper bound vectors:

$$\mathbf{L} := \mathbf{y}^0_{\text{post}} + \bar{\varepsilon} \cdot \mathbf{1} - c_\alpha \cdot \mathbf{1}, \quad \mathbf{U} := \mathbf{y}^0_{\text{post}} + \bar{\varepsilon} \cdot \mathbf{1} + c_\alpha \cdot \mathbf{1}, \tag{3.21}$$

which define the conformal confidence band around the counterfactual trajectory.

This approach allows finite-sample valid coverage under the assumption that the pre- and post-treatment residuals are exchangeable. Intuitively, if the SHC model fits the pre-treatment series well (i.e., residuals are small and centered), then its extrapolations into the post-treatment period are expected to carry similarly bounded uncertainty. In this sense, the residual distribution learned from the evaluation window is used to calibrate uncertainty for the out-of-sample outcomes.

These prediction intervals reflect uncertainty in the estimated counterfactual. Hence, if the observed outcome falls far outside the prediction interval, this is interpreted as evidence of a treatment effect. Unlike traditional confidence intervals around treatment effect estimates (e.g., using standard errors), the conformal intervals here do not assume a data-generating model. I use a default coverage level of 90% (i.e., $\alpha = 0.1$) in all applications.

4. DATA

To evaluate the impact of Hawaii's COVID-19 border closure on its tourism economy, I use monthly time series spanning multiple economic dimensions, focusing on four primary outcomes: two related to tourism demand and two capturing broader economic impacts. All outcomes are transformed into annualized year-over-year (YoY) growth rates. This transformation mitigates nonstationary trends—crucial for the validity of the SHC estimator Chen et al. (2024)—and allows treatment effects to be interpreted directly as percentage point changes.

The first two outcomes capture the demand side of Hawaii's tourism sector. Visitor Days measures the total number of person-days spent by visitors in a given month, integrating both the volume of arrivals and their length of stay. Occupancy refers to the statewide hotel occupancy rate, defined as the percentage of available hotel rooms that are occupied. Together, these indicators provide complementary measures of tourism demand: the former captures total visitor presence, while the latter reflects realized capacity utilization across the hospitality sector. Both series are sourced from Hawaii's Department of Business, Economic Development & Tourism (DBEDT) and span from January 1990 to December 2020.

The remaining outcomes capture economic effects, one sector-specific and one broader in scope. *Accommodation Employment* tracks the number of employees in Hawaii's accommodation industry, a labor-intensive sector directly tied to tourism activity. *The Coincident Economic Activity Index* summarizes statewide macroeconomic conditions based on four indicators: nonfarm payroll employment, the unemployment rate, average weekly hours worked in manufacturing, and wage and salary disbursements. This index is benchmarked to match the trend of gross state product, making it a comprehensive indicator of the state's economic health. Both economic indicators are sourced from the Federal Reserve Economic Data (FRED) system and cover January 1991 to December 2020.

Hawaii's mandatory quarantine policy took effect in late March 2020. I define the intervention to begin in March 2020, corresponding to time $t = 362$, and formalize the treatment indicator as $d_t := \mathbf{1}\{t \geq 362\}$, so that $d_t = 1$ for all months after the policy's implementation and $d_t = 0$ otherwise. The final pre-treatment period is February 2020, yielding a pre-treatment sample of $T_0 = 361$ months.

5. RESULTS

In Figure 1 we visualize the outcomes of interest over time from January of 1991 to December 2020. We can see that the growth rate trends for all outcomes were relatively smooth across the full pre-intervention series, exhibiting seasonality and relatively low variance. We also see that the outcomes of interest precipitously dropped when the mandatory quarantine policy was implemented. The plot also suggests that Assumption 3.1 may be valid. For example, for Assumption 3.1 to be violated, we would need a situation where the outcome had extremely high variance. While we see seasonality across all outcomes, the time series seemed smooth enough to be approximated by pre-treatment observations.

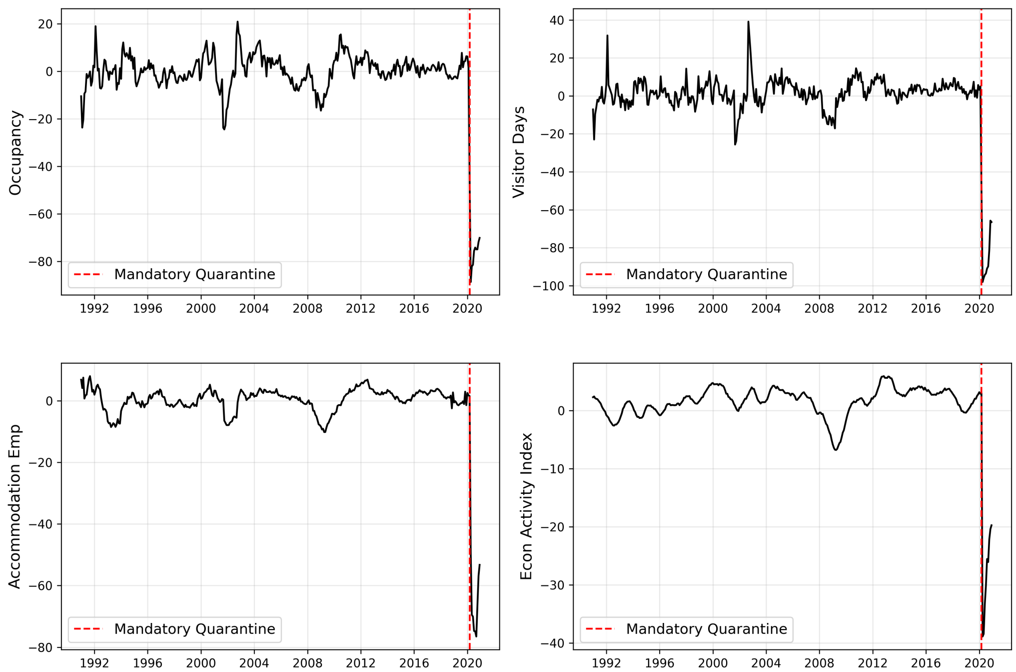

Figure 1: Outcome Trends.

Figure 2 presents the initial SHC results. For these results, $m = 60$ (5 years of evaluation data), but the main results remain robust up until 8 years of evaluation data, with slight deviations in uncertainty.

Across all four primary outcomes, I find large and precisely estimated negative effects from Hawaii's border closure. Note that the good in-sample fit suggests that Assumption 3.2 is satisfied. Tourism demand was the most directly affected: Visitor Days declined by 85.4 percentage points, and Hotel Occupancy fell by 75.8 percentage points, relative to their synthetic historical counterfactuals. These are dramatic, near-total collapses in the state's primary economic driver. The SHC model achieves a close pre-treatment fit for both outcomes (RMSEs ≈ 1), suggesting that the model is well-calibrated to detect meaningful post-treatment divergence.

Labor market and broader economic indicators reflect these shocks. Accommodation employment declined by 60.7 percentage points, while the Economic Activity Index fell by 27.7 percentage points. Though these outcomes are more aggregated, they follow the same pattern as the tourism indicators—abrupt declines followed by a gradual recovery

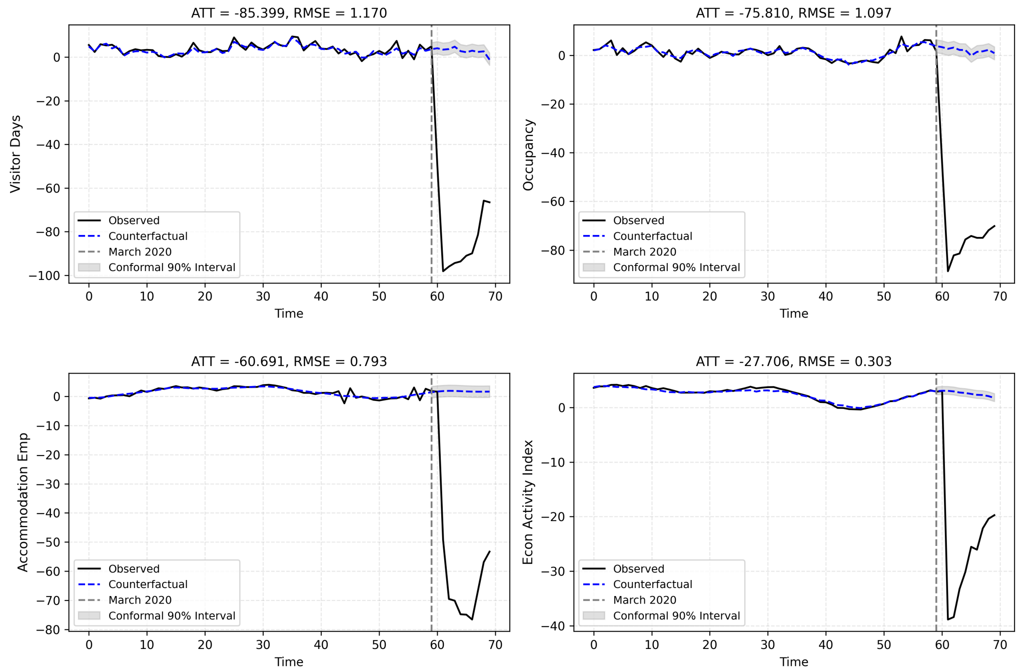

Figure 2: SHC Analysis.

towards the end of 2020, suggesting Hawaii's shutdown sent a measurable shock through the broader economy. Notably, in all four cases, the observed values after March 2020 fall well outside the 90% conformal prediction intervals, strengthening the inference that these declines reflect true treatment effects rather than model noise or volatility.

The results also suggest the mechanics of the causal effect: the sharp decline in overall employment, especially in accommodations, is plausibly driven by the collapse in tourism. Given Hawaii's economic structure, the vast majority of hotel/hospitality employment is directly demand-driven. Thus, the drop in visitor-days and occupancy rates leads to reduced staffing needs across the sector. This channel provides a natural explanation for the labor market contraction observed in the data. Importantly, the pattern across all outcomes suggests a consistent causal chain: border closure leads to a collapse in tourism, which in turn leads to employment contraction, macroeconomic slowdown, and a fiscal shock. This sequential relationship reinforces the credibility of the estimates and gives confidence that the economic effects documented here reflect real-world consequences of the policy intervention.

Robustness checks support this interpretation. Using 10 years of pre-treatment data produces nearly identical ATT estimates, with only minor degradation in early-period fit. The central pattern—a sharp short-run shock followed by recovery—holds across specifications. The results are presented in Figure 3.

Placebo tests using a September 2019 "fake" intervention date yield negligible effects, with estimated ATTs well within the model's baseline error margins. This provides re-

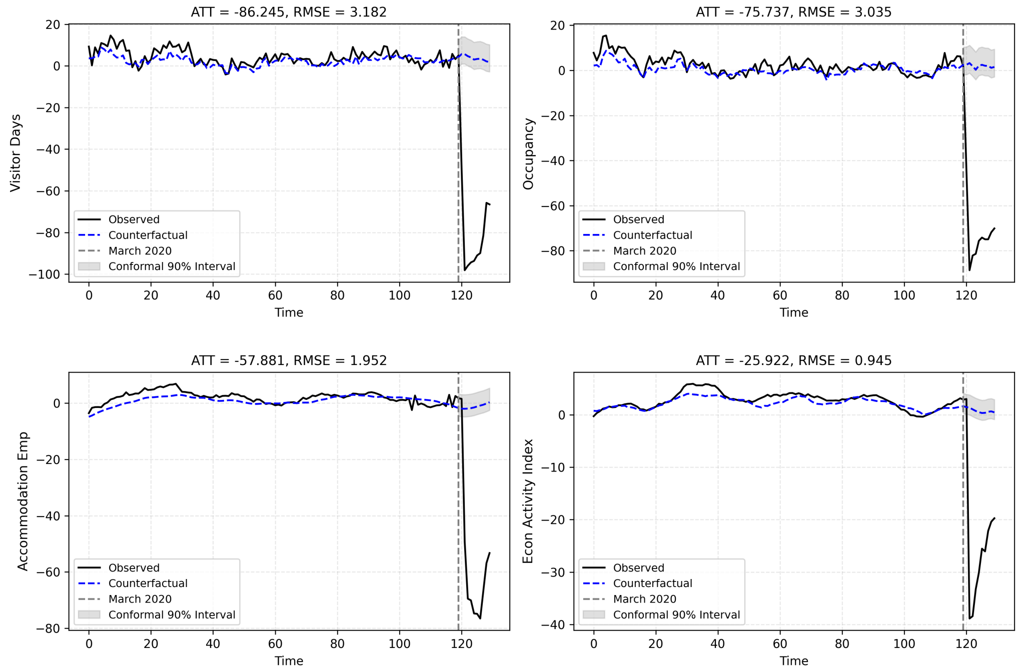

Figure 3: SHC Analysis, $m = 120$.

assurance that the model is not spuriously detecting treatment effects where none exist. Figure 4 presents these results.

These results offer two key takeaways. First, the SHC model does not erroneously detect treatment effects when none exist—it exhibits strong temporal specificity. Second, even in cases where placebo ATTs are slightly nonzero, they fall within normal forecast uncertainty. This is confirmed by the conformal prediction intervals, which entirely cover the observed outcomes during the placebo window. That is, any apparent deviation between predicted and realized values is consistent with random variation under no treatment. These placebo checks strengthen the case for a causal interpretation of the post-March 2020 findings: the observed deviations are not artifacts of overfitting or model instability, but genuine responses to Hawaii's border closure.

To estimate the impact of the mandatory quarantine on fiscal losses, I collect data on taxes generated from Hawaii's General Excise tax for the hotel industry and overall transient accommodations, using quarterly revenue data from 2010 to 2019 (with the pre-treatment period being from 1991 to the end of 2009). I estimate a 91.7 percentage point decline in Transient Accommodations Tax receipts, and an 88.3 point drop in hotel-related General Excise Taxes, consistent with large-scale fiscal losses during the quarantine period. These results are presented in Figure 5.

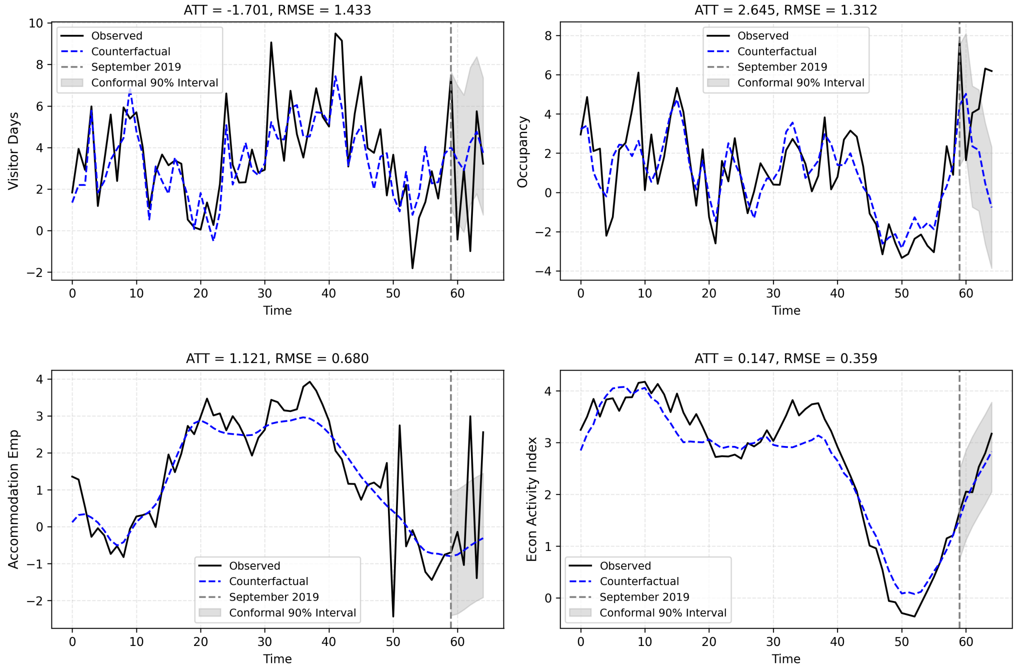

Figure 4: Placebo Studies.

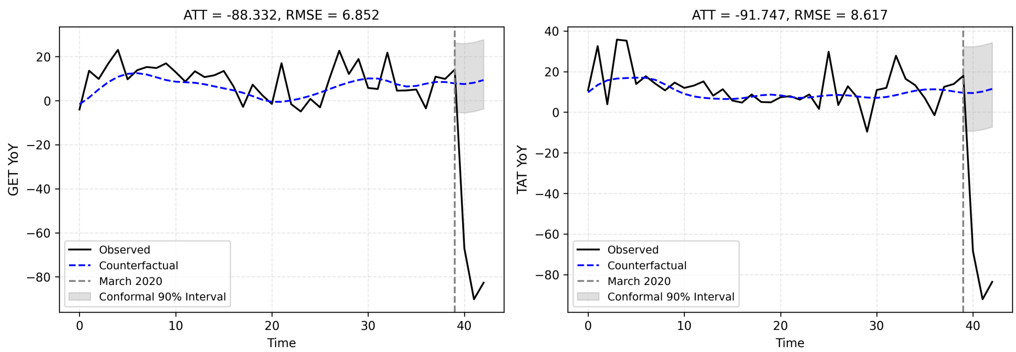

Figure 5: Additional Outcomes.

6. DISCUSSION

This paper provides a quantitative estimate of the short-run economic costs of an early, aggressive pandemic response. Hawaii's mandatory quarantine policy caused a near-total collapse in tourism, with sharp but largely temporary contractions in employment and broader economic activity. Despite its severity, these effects were not persistent: most metrics rebounded by early 2021. This pattern—steep initial losses followed by gradual recovery—suggests that decisive early action may impose short-run pain but help avoid longer-term disruption.

These results contribute to an ongoing debate over whether early containment policies are worth their economic cost. Hawaii's case supports the theory of "front-loaded sacrifice": by absorbing short-term damage, the state avoided sustained epidemiological and economic chaos. This interpretation echoes findings from Ke and Hsiao (2022) on Hubei's lockdown and aligns with the intuition of models emphasizing early interventions to delay or avoid exponential harm.

Bond-Smith and Fuleky (2022) provide useful context. They document that, as of 2022, Hawaii's non-farm employment remained nearly 10% below pre-pandemic levels, raising concerns about persistent scarring. Yet they also note that Hawaii avoided major COVID waves until Delta and Omicron. This contrast—between early economic contraction and delayed epidemiological impact—underscores the tradeoff central to this paper: early action can buy time, but at an economic cost.

This study does not estimate public health benefits directly, nor does it claim Hawaii's policy was cost-effective overall. However, it does help clarify the economic price of acting early. Future research should evaluate whether these costs were offset by reduced mortality, hospital strain, or faster recovery. More broadly, Hawaii illustrates the challenge of crisis policymaking under uncertainty.

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